Case Study 3

HES 3000-002: Sport Performance Analytics

Fall 2026

Overview

In this activity, you will use team-level box score data from the 2025–26 Power conferences to investigate relationships between performance metrics. You will calculate and interpret correlation coefficients, create scatter plots with linear trend lines, and use the trendline equation to make predictions

Part 1: Formulate Your Question & Identify Variables

Before beginning the analysis, look through the column headers of your dataset (power_conferences_2526_summary.xlsx) and think about basketball strategy. Formulate a research question that tests a logical relationship between two numeric variables (e.g., "Does a team's average 3-point attempt volume have a positive linear relationship with their scoring output?" or "How do turnovers relate to offensive scoring averages?").

Answer the following for Part 1:

- State what one row of the dataset represents.
- Write out your research question.
- Identify the predictor variable (x) and outcome variable (y) by their exact column names.
- Define what each variable measures and briefly explain why you expect a relationship.

Part 2: Run the Correlation Analysis

Measure the strength and direction of the linear relationship between your two chosen variables. This can be done with Excel's =CORREL() function.

Report the following for Part 2:

- Exact *r* value generated by Excel.
- What does r tell you about the direction and strength of the linear relationship between your variables?

Part 3: Create the Scatter Plot & Add a Trend Line

As discussed in class, visualizing the data helps us check for patterns, spread, and outliers. Create a scatter plot of your variables of interest. **Clearly title the plot and label your axes**. (Note: The predictor variable should be on the x axis and the outcome variable on the y axis.)

Add a linear trendline and answer the following:

- **Does a linear trendline appear appropriate for this relationship?** Explain your answer using the pattern in / shape of the scatterplot.
- If you think the line is a poor fit, **explain how this affects the usefulness of your prediction** in Part 5.

Part 4: Extract the Equation and Interpret Your Findings

To use your model for estimation, you need the line's mathematical formula. Get the equation for your linear trendline and answer the following:

- Look at the slope (*m*) in your chart equation. In practical basketball terms, **what does this slope mean for every 1-unit increase in your *x* variable?**
- Look at the y-intercept (b) in your chart equation. In practical basketball terms, **what does this intercept mean? Does this make sense to interpret for your research question?**

Part 5: Prediction

Use your chart's linear equation ($y = mx + b$) to estimate an outcome. For part 5, pick a realistic value for your x variable that falls within the range of your dataset and solve for y. Then, answer the following:

- What value did you select as your input?
- Show your equation, the substitution of your chosen x value, and the predicted y value with units.
- Write a short interpretation of what your model predicts.

Bonus question: Why might a team's actual outcome differ from this prediction?

Every statistical model has error (residuals). Actual team performances don't always sit perfectly on your trend line.

- Identify one team whose actual y value is noticeably higher or lower than the model predicts.
- What are two other factors or variables that your model does not account for that could help explain this team's performance? Briefly explain how each might contribute to the difference between its actual and predicted performance.